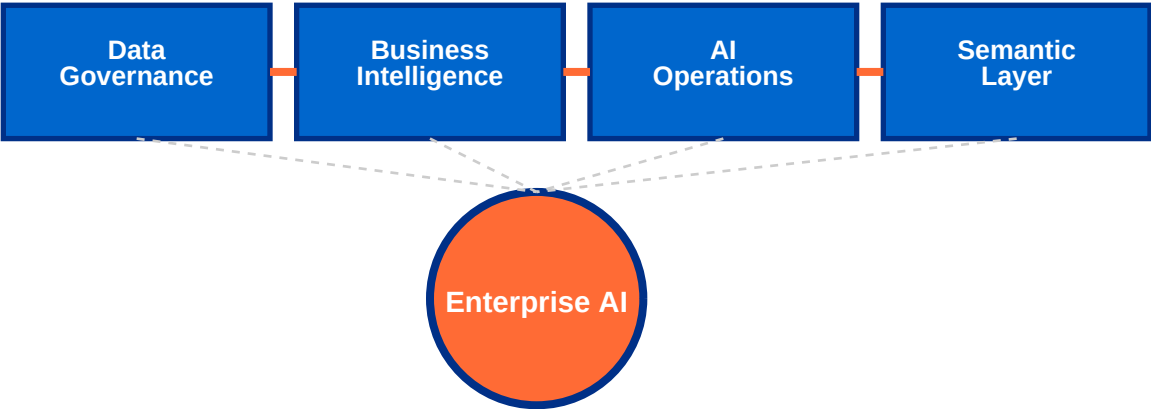




AI & Governance Blueprint

For Yaskawa America

*A Comprehensive Framework for Enterprise AI Integration
with SAP ERP/CRM Systems and Robust Governance*



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Owner:	Chief Data Officer

Executive Summary

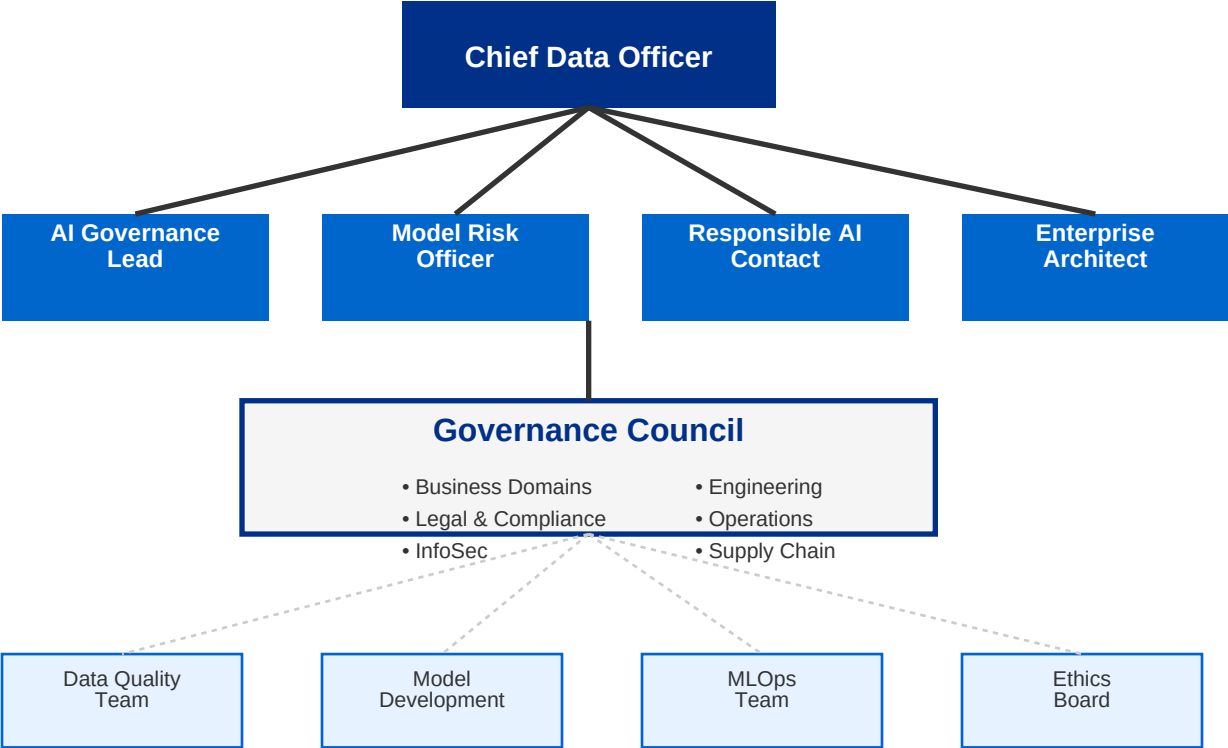
This blueprint provides a comprehensive framework for Yaskawa America to formalize AI across the organization, integrating deeply with SAP ERP/CRM systems while establishing robust governance across data, business intelligence, semantics, and AI operations. Drawing from enterprise best practices, this plan addresses AI integration across product lifecycle management, engineering operations, and supply chain systems.

- Centralized AI Governance
- Multi-Layered Framework
- SAP Integration
- Responsible AI
- Continuous Monitoring

Strategic Value Proposition

Dimension	Value
Operational Excellence	Predictive maintenance, optimized production scheduling, quality improvement
Customer Experience	AI-powered support, intelligent case routing, personalized service
Innovation Acceleration	Generative design, rapid prototyping, market intelligence
Risk Mitigation	Compliance automation, bias detection, robust governance
Cost Optimization	Inventory reduction, energy efficiency, resource allocation

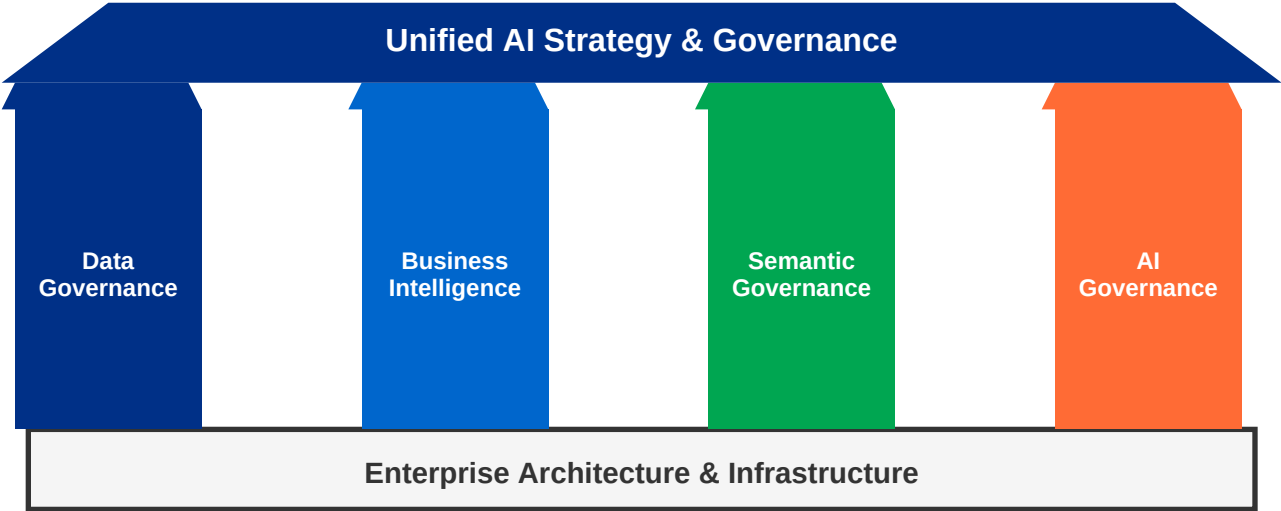
1. AI & Data Organization Structure



Leadership Roles & Responsibilities

Role	Key Responsibilities
Chief Data Officer	Overall AI and data strategy, executive sponsorship, resource allocation, stakeholder alignment
AI Governance Lead	Policy enforcement, compliance monitoring, framework implementation, audit coordination
Model Risk Officer	Risk assessment, validation protocols, model performance monitoring, escalation management
Responsible AI Contact	Ethics oversight, bias detection, fairness monitoring, policy development
Enterprise Architect	Technical integration, infrastructure design, SAP connectivity, platform selection

2. Multi-Layered Governance Framework



2.1 Data Governance

Implementing a Data Mesh architecture to break down traditional silos across SAP modules (MM, PP, SD, QM, PM) with domain-oriented ownership and self-serve infrastructure.

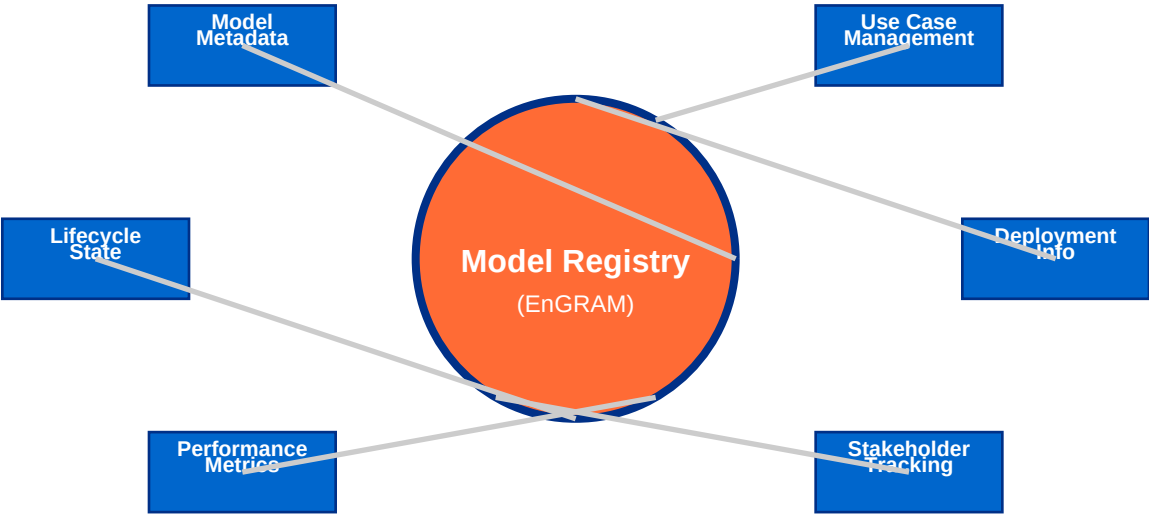
Level	Description	Examples
Public	Publicly available information	Product brochures, press releases
Internal Use	General business information	Internal reports, process docs
Confidential	Sensitive business data	Customer data, pricing, specifications
Restricted	Highest sensitivity	IP, trade secrets, competitive intelligence

Data Quality Framework:

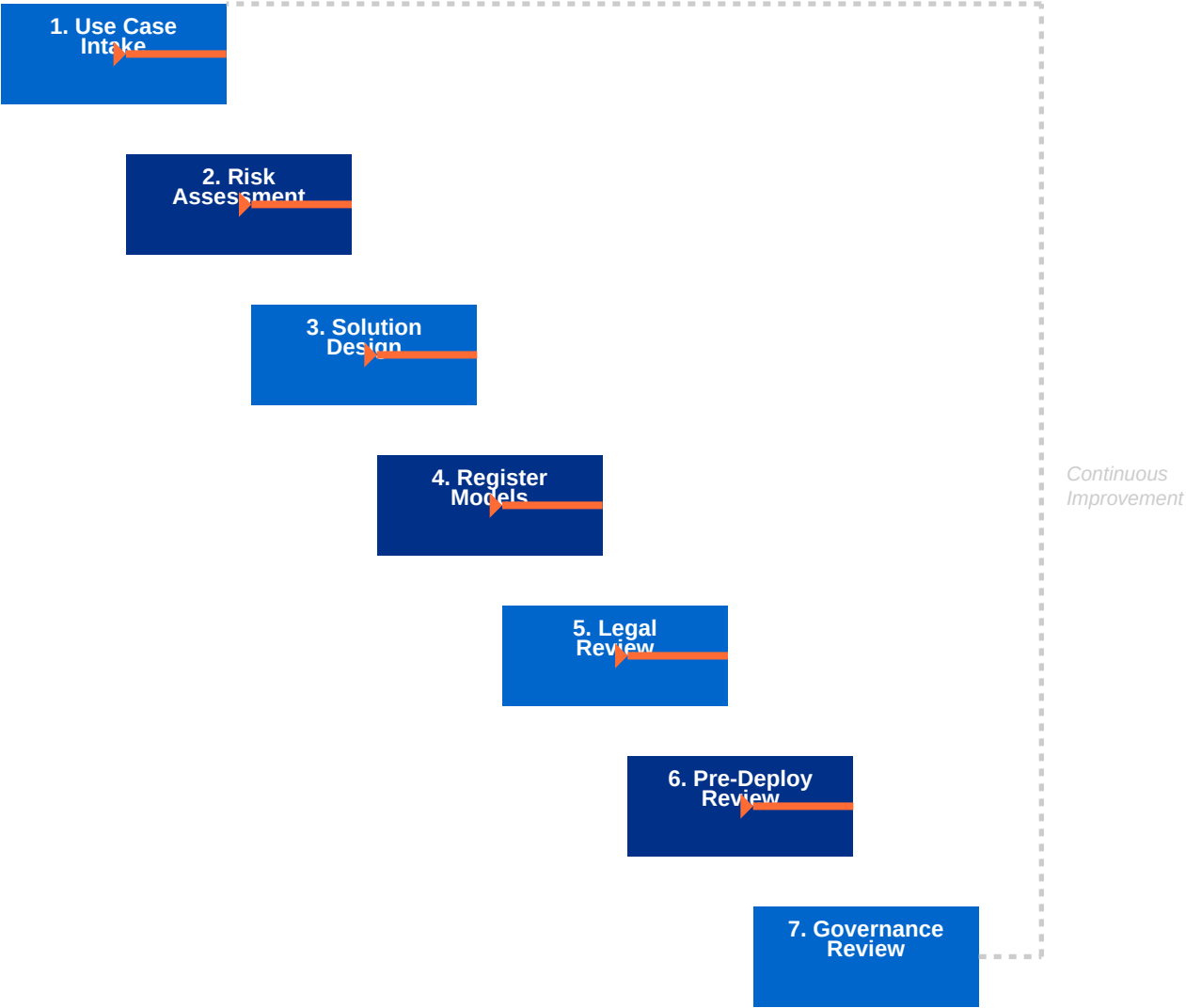
- Accuracy:** Validation of data correctness against source systems
- Completeness:** Checks for missing or null values in critical fields
- Timeliness:** Monitoring data freshness and update frequency
- Consistency:** Alignment across SAP and external systems
- Validity:** Rules aligned with engineering specifications

2.4 AI Governance

Implementing an enterprise governance registry for administration of models (EnGRAM-style) with comprehensive tracking of model metadata, use cases, and stakeholder accountability.



3. Enhanced Use Case Lifecycle Phases

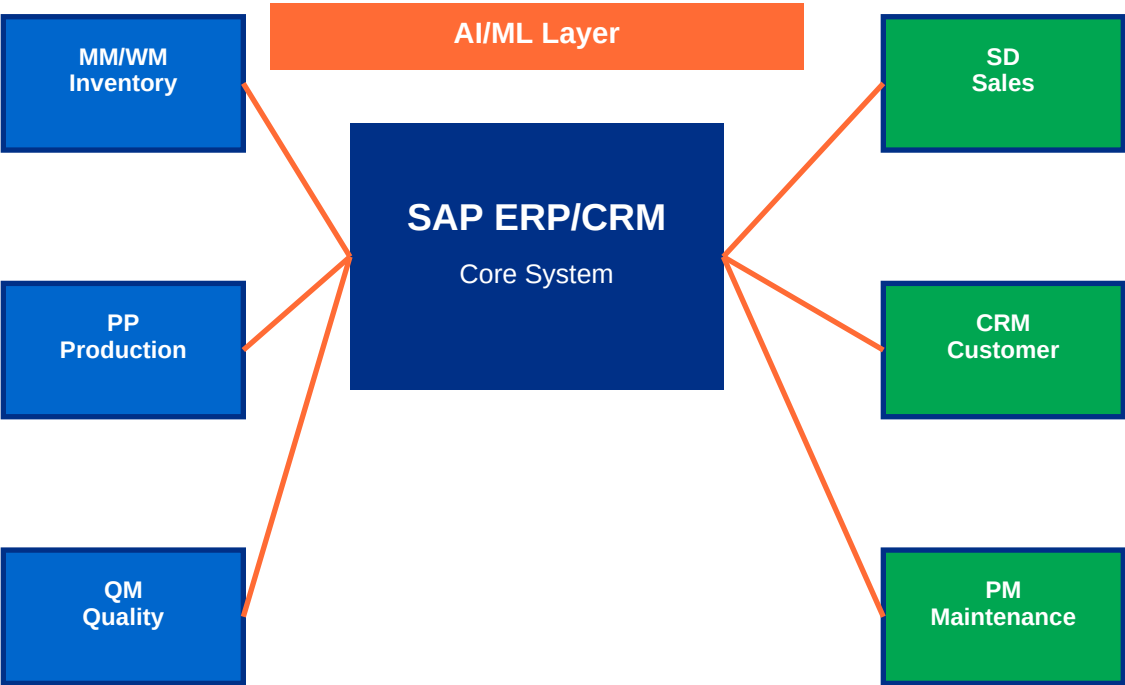


Phase Details & Key Activities

Phase	Key Activities	Deliverables
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Use Case Intake	• Business problem identification • Feasibility assessment • Strategic alignment • Resource estimation	Use case charter, initial ROI estimate
Risk Assessment	• Usage risk scoring • Data access risk • Model complexity evaluation • Material model determination	Risk scorecard, mitigation plan
Solution Design	• Architecture blueprint • SAP integration points • Data pipeline design • Infrastructure requirements	Technical design document, integration specs
Register Models	• Model documentation • Training data cataloging • Bias assessment • Feature engineering	Model registry entry, data lineage
Legal Review	• IP considerations • Data usage rights • Third-party licensing • Export control compliance	Legal approval, contract amendments
Pre-Deploy Review	• Model validation • Performance testing • Security assessment • UAT completion	Validation report, test results
Governance Review	• Stakeholder sign-off • Go-live approval • Monitoring activation • Documentation check	Governance approval, deployment plan

4. AI Integration Across Key Business Areas



4.1 SAP ERP Integration - Internal Operations

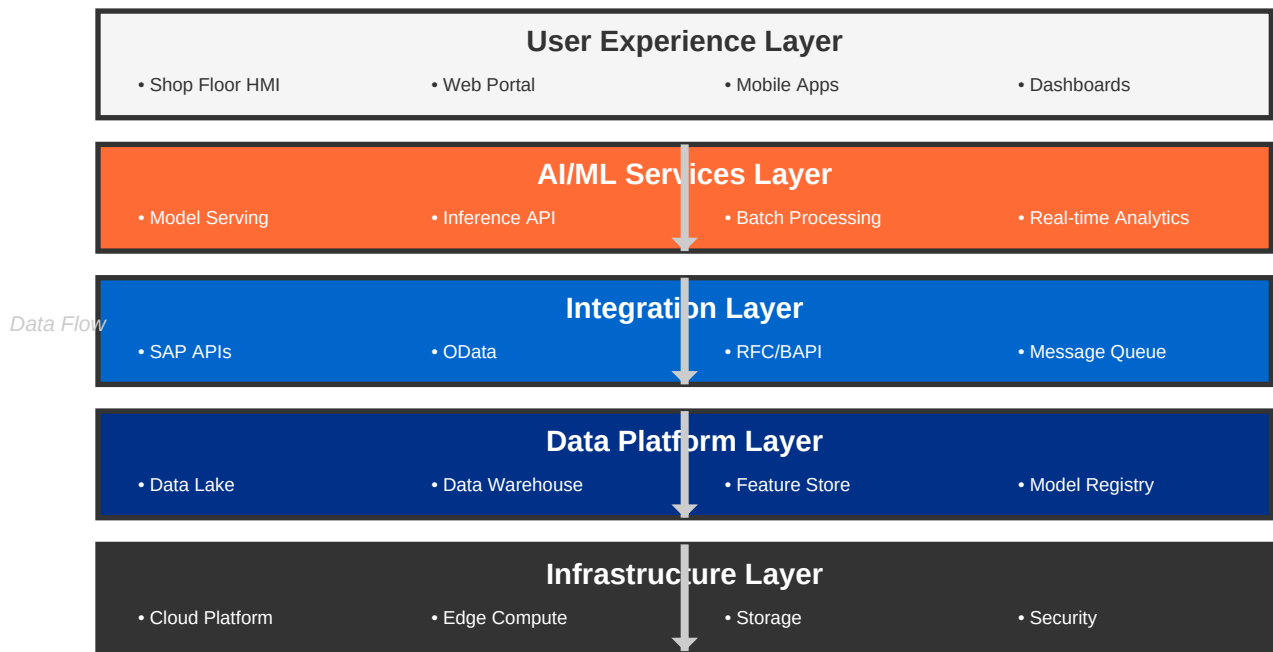
SAP Module	AI Use Cases	Business Impact
PM - Predictive Maintenance	• Real-time sensor data analysis • Automatic work order generation • Spare parts forecasting • Equipment health scoring	Reduced downtime, optimized maintenance costs, improved equipment reliability
PP - Production Planning	• AI-driven scheduling optimization • Capacity planning with ML • Bottleneck prediction • Dynamic routing	Increased throughput, better resource utilization, faster delivery
QM - Quality Management	• Computer vision inspection • Defect pattern recognition • Predictive quality alerts • Supplier quality scoring	Reduced defects, improved customer satisfaction, lower rework costs

MM/WM - Inventory	• Demand forecasting • Safety stock optimization • Reorder point automation • Warehouse optimization	Lower inventory costs, reduced stockouts, improved cash flow
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4.2 SAP CRM Integration - Customer Experience

- Virtual Assistant:** NLP-powered chatbot for technical support queries with 24/7 availability
- Case Routing:** Intelligent assignment based on technician expertise, availability, and customer priority
- Solution Recommendation:** ML-driven troubleshooting from knowledge base with contextual suggestions
- Agent Assist:** Real-time guidance with access to product manuals, service bulletins, and parts info
- Sentiment Analysis:** Monitor customer satisfaction, trigger escalation when needed
- Lead Scoring:** Prioritize opportunities based on propensity to convert and customer value

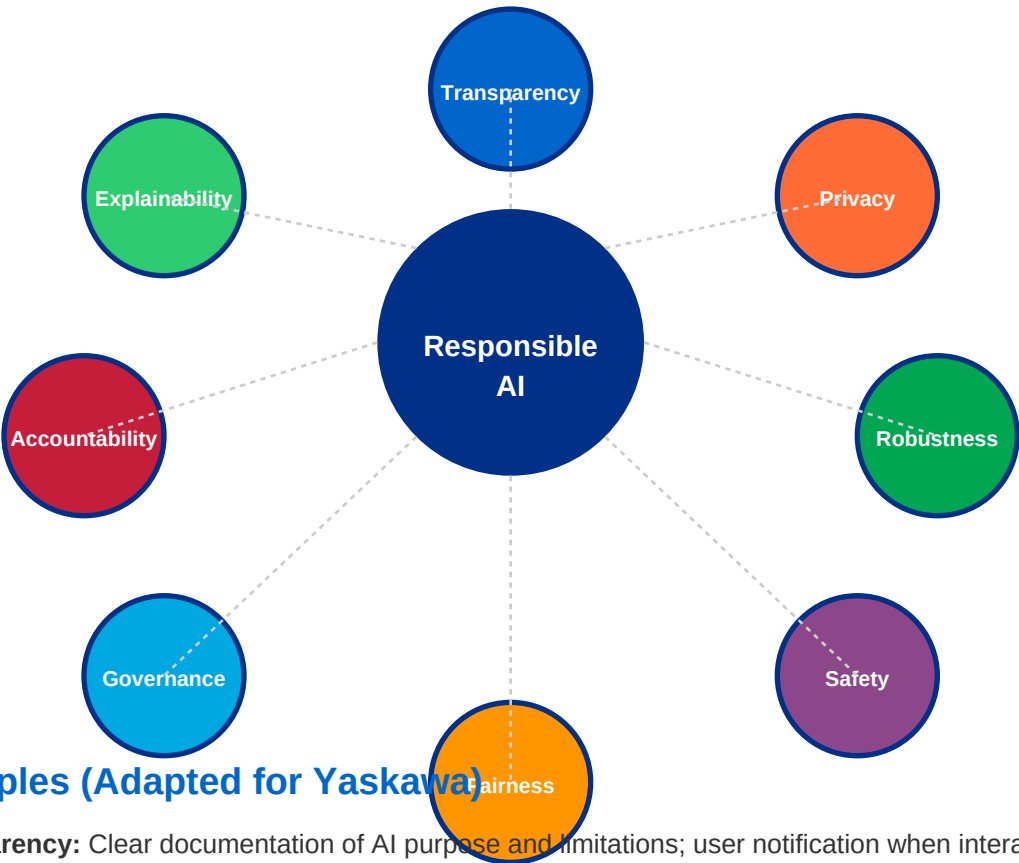
5. Technical Architecture & Infrastructure



Infrastructure Components

Component	Purpose	Technologies
Edge Computing	Real-time AI inference at manufacturing sites	NVIDIA Jetson, Azure IoT Edge, Local GPUs
Cloud Platform	Centralized training, development, and serving	Azure ML, AWS SageMaker, or GCP Vertex AI
SAP Integration	Bidirectional data exchange with ERP/CRM	SAP BTP, OData, RFC/BAPI, MuleSoft
Data Lake	Historical data storage for analytics and training	Azure Data Lake, S3, Databricks
Security Layer	Authentication, authorization, encryption	OAuth 2.0, RBAC, TLS/SSL, Azure AD

7. Responsible AI Framework



AI Principles (Adapted for Yaskawa)

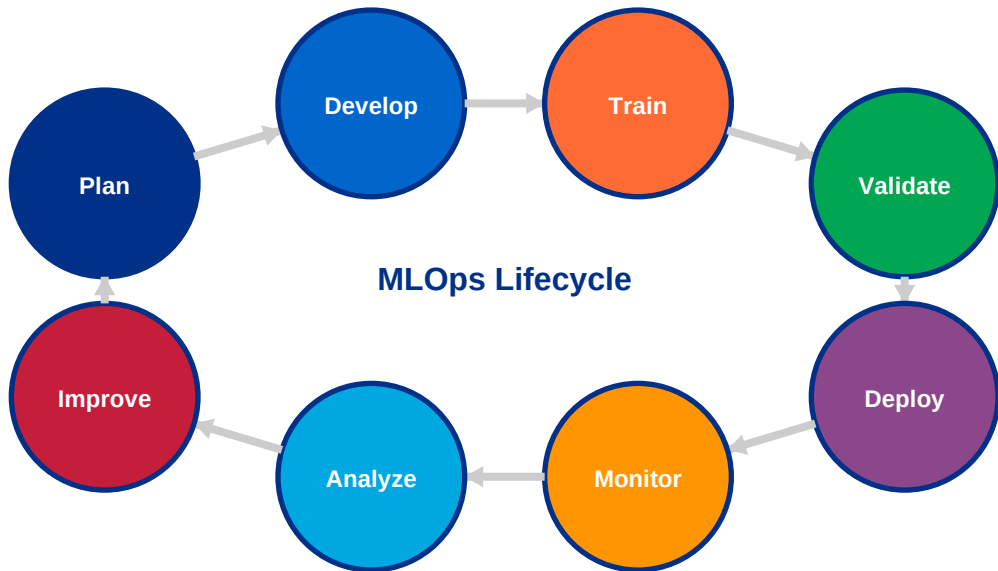
- Transparency:** Clear documentation of AI purpose and limitations; user notification when interacting with AI; explainable decisions for critical processes
- Privacy:** Data minimization principles; customer/supplier data protection; employee privacy in workplace AI; anonymization where possible
- Robustness:** Rigorous testing before deployment; continuous monitoring for drift; fallback mechanisms for failures; regular revalidation cycles
- Safety:** Human-in-the-loop for high-stakes decisions; override capabilities for automated systems; fail-safe designs; emergency stop procedures
- Fairness:** Bias detection and mitigation; fair treatment in HR applications; equitable customer service; supplier diversity considerations
- Governance:** Clear accountability structures; approval workflows; audit trails; regular governance reviews

Model Validation Framework

Validation Type	Focus Areas	Methods
Conceptual	Theory soundness, methodology appropriateness, use case relevance	Peer review, literature comparison, expert assessment

Implementation	Code quality, data processing accuracy, output correctness	Code reviews, unit testing, integration testing
Performance	Accuracy metrics, robustness, latency, resource usage	Backtesting, stress testing, benchmark comparison
Ongoing	Drift monitoring, bias detection, performance tracking	Statistical tests, monitoring dashboards, periodic audits

8. Continuous Monitoring & MLOps



Model Performance Monitoring

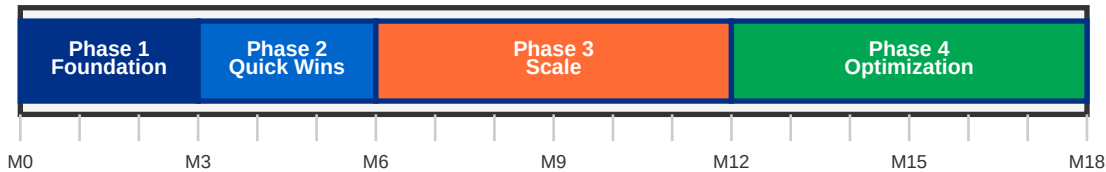
Metric Category	Specific Metrics	Alert Thresholds
Technical Metrics	Prediction accuracy, inference latency, resource consumption	Accuracy < 95%, latency > 50ms, errors > 1%
Business Metrics	Impact on KPIs (OEE, quality, cost), user adoption, ROI	ROI decline > 10%, adoption < 50%, negative ROI
Data Quality	Distribution shifts, missing values, outliers, feature correlations	Drift score > 0.3, missing > 5%, outliers > 2%
Model Health	Confidence scores, prediction diversity, output consistency	Low confidence > 20%, low diversity detected

Model Revalidation Schedule

- Quarterly Reviews:** Performance against benchmarks, data quality assessment, stakeholder feedback, compliance check
- Annual Comprehensive Audit:** Full validation cycle, updated risk assessment, documentation update, strategic alignment review

Trigger-Based Revalidation: Significant performance degradation, major process changes, regulatory updates, significant data drift

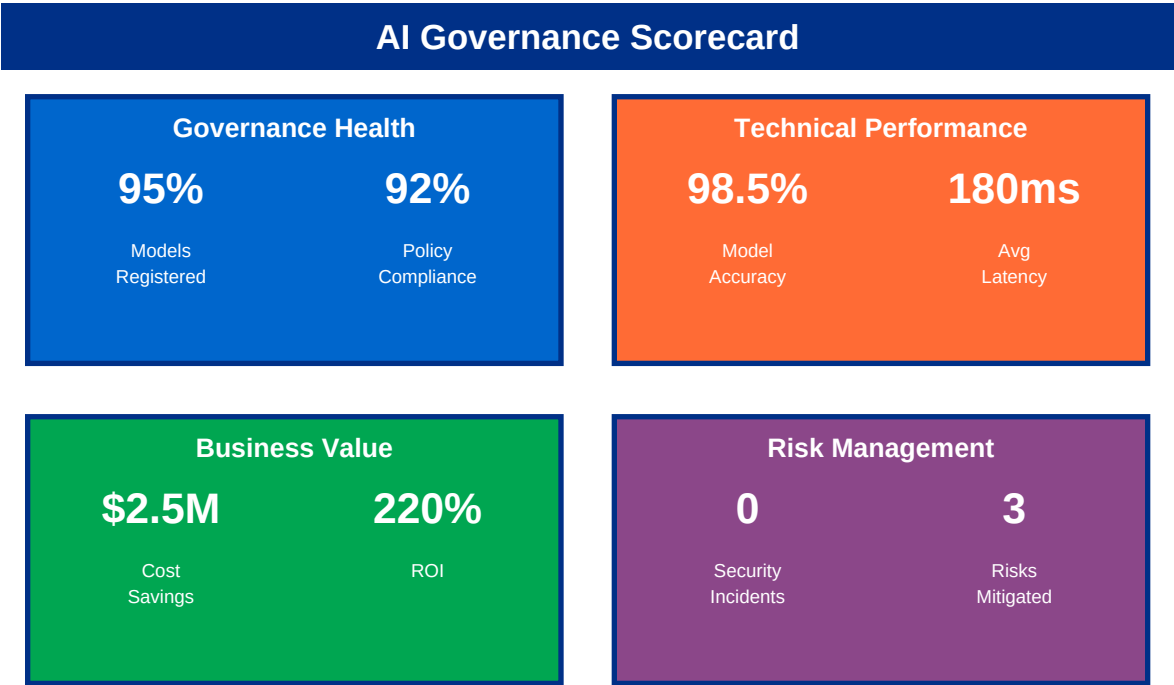
12. Implementation Roadmap



Phased Implementation Approach

Phase	Timeline	Key Deliverables	Success Criteria
Foundation	Months 1-3	- AI Governance Office established - Model registry implemented - Data classification complete - Initial use cases identified	Governance structure operational, 5+ use cases in pipeline
Quick Wins	Months 4-6	2-3 pilot AI models deployed - SAP integration patterns - Knowledge repository foundation - User training programs	3 models in production, positive user feedback, measurable
Scale	Months 7-12	- Expanded use cases across domains - AI development platform deployed - Advanced MLOps practices - Self-service analytics	10+ models in production, governance compliance >95%
Optimization	Months 13-18	- Model refinement and tuning - Third-party AI partnerships - Advanced capabilities (GenAI) - Center of excellence	ROI >200%, industry recognition, knowledge sharing active

13. Success Metrics & KPIs



Comprehensive KPI Framework

Category	Key Performance Indicators	Target
Governance	% of AI models registered, avg approval time, compliance rate, audit score	>95%, <12 weeks, >90%, >4.0/5
Technical	Model accuracy vs. baseline, inference latency, system uptime, data quality score	>90%, <500ms, >99.5%, >85%
Business	ROI by use case, cost savings, efficiency gains, customer satisfaction	>150%, \$1M+/year, 20%+, >4.2/5
Risk	Security incidents, bias incidents, model failures, regulatory compliance	0 incidents, 100%

Conclusion

This blueprint provides Yaskawa America with a comprehensive, structured approach to AI implementation that balances innovation with governance. By learning from enterprise best practices while tailoring to manufacturing and robotics industry needs, this framework ensures responsible, scalable, and value-driven AI adoption across the organization.

Keys to Success

- 1 Executive Sponsorship
- 2 Cross-Functional Collaboration
- 3 Incremental Implementation
- 4 Continuous Learning
- 5 Cultural Transformation

Next Steps: Form the AI Governance Office, identify executive sponsor, and begin Phase 1 foundation activities. Schedule kickoff meeting with key stakeholders within 30 days.

For More Information	
AI Governance Office	ai-governance@yaskawa.com
Chief Data Officer	cdo@yaskawa.com
Document Repository	Internal SharePoint: AI Governance Hub